

“RAS (Rubber Agroforestry Systems) and Climate Change in South East Asia : adaptation, mitigation, contribution ?

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Rationale of the FORSEA project

- Lack of knowledge on the behavior (survival, growth, production...) of the rubber tree under high temperatures coupled with a high atmospheric CO₂,
→ as the rubber tree has never been cultivated in areas where annual mean temperature is above 28°.
- Forecasted increase of climate extreme events frequency: drought, storms, flooding, heat waves, likely impacting the rubber production.
- Uncertainty regarding the future availability of agricultural manpower/tapping workforce in rubber plantations (companies and smallholdings).
- Which **adaptation** for RAS to these evolutions.

→ **Workpackage WP3 : Socio-economics and Labor Evolution Scenarios**

- Social economic context of RAS development and Labor Availability

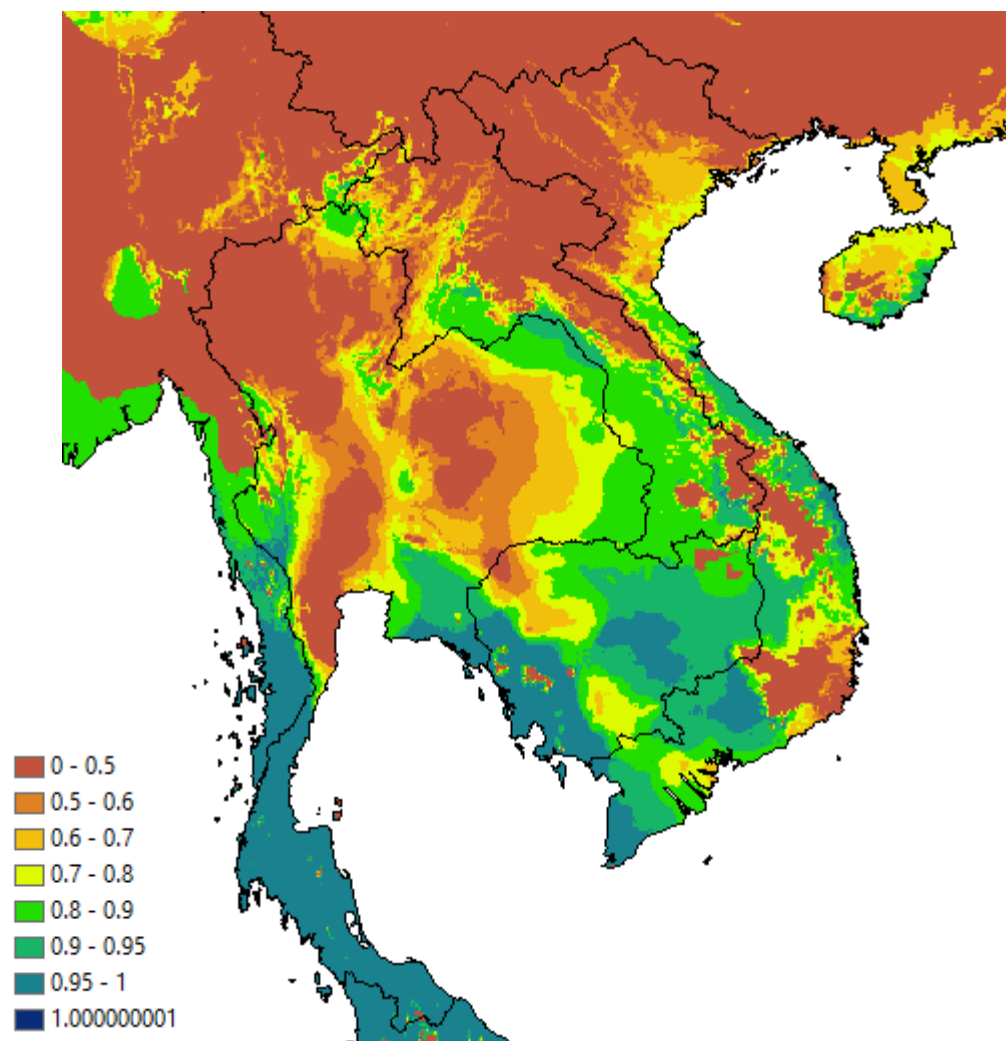
→ **Workpackage WP4 : Agronomy and adaptation scenarios**

- Adaptation and Innovations in Response to new contexts in the very next future

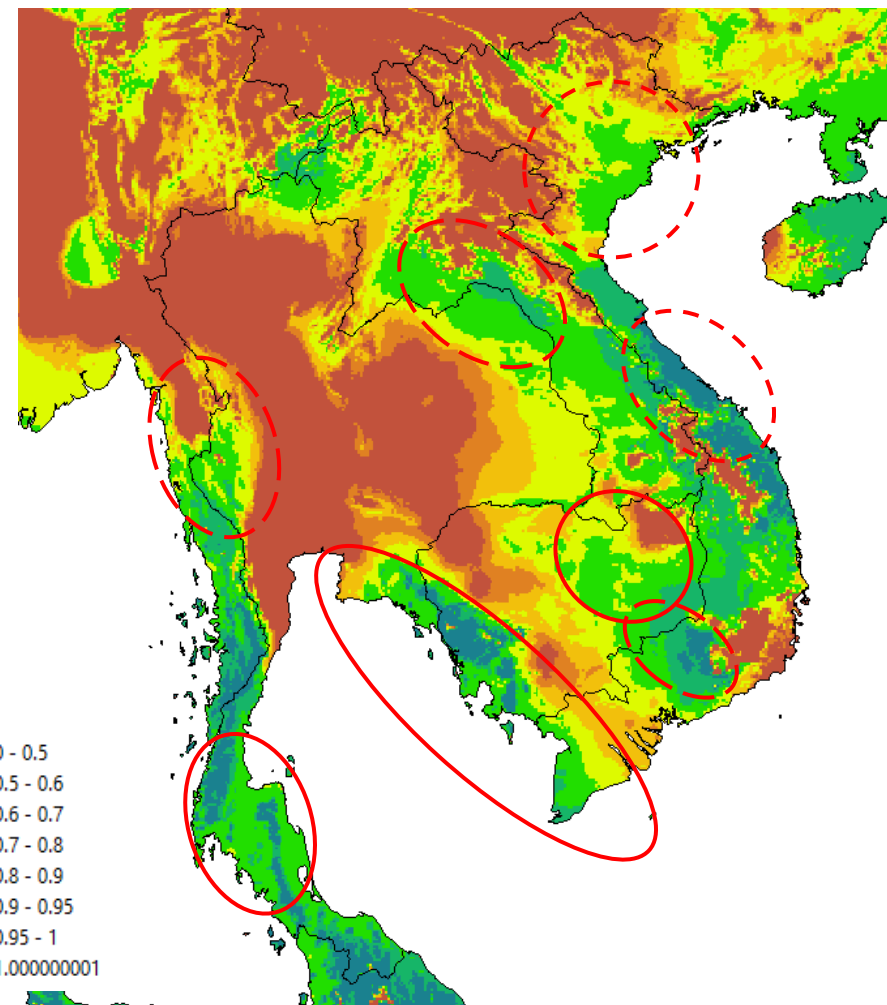
To address the following challenges:

- ✓ What will be the behavior of the rubber tree under climate change scenarios within 10-30 years (and more) ?
- ✓ And for RAS rubber and associated trees ?
- ✓ Who will tap the rubber trees within 10-30 years in a context of economic and academic development (agriculture / industry and services)
- ✓ ? Competition with other crops.... Oil palm
- ✓ What are the possibilities of simultaneous adaptation to these 2 global changes?

Baseline, 2010-2019



Mean from the results of 21 climate models, 2-4.5, 2081-2100



- Increase
- Stability
- Decrease

What is at stake : adaptation of RAS, effective mitigation/contribution of RAS to CC ????

- CC evolution can limit the expansion of suitability areas : moving from « traditionnal » to « marginal »
- The high temperature ($>28^{\circ}\text{C}$) index is hypothetical. More research is needed to see what is the real temperature limit
- → can RAS with more biomass and biodiversity create conditions for a better adaptation ? In terms of soil, water catchment, canopy temperature ?
- → more biomass = more competition ?
- Currently : we have hypothesis and plausible expectations on some RAS patterns to mitigate impact of CC
- → but no physical data to prove it : → we need trials and observations
- → and which RAS patterns are the most adapted ?

CC current existing impact in some countries

Myanmar : 6 months with heavy rainfall and 6 « dry » months : potential significative competition

Cambodia : 2 to 5 dry months depending on location and soils

Sri Lanka/Eastern area : idem to Cambodia

Indonesia/South Sumatra : 4/5 dry months now (2 months before)

China/Hainan and Southern Yunan : less global rainfall....

Vietnam/highlands : altitude and dry season

Thailand/Northeast : poor sandy soils and potentially crops with heavy competition for water (cassava..

Current impact of CC

Indonesia

Significant increasing of leaf diseases pressure

Poor tapping and D1

Rubber production decreases from 1200 kg/ha/year in 2000 to 650/850 kg/ha/year in 2024

No more replantation since 2019

RAS with selected trees that requires lee water : timber, gaharu (*Aquilaria spp*).....

Sri lanka

High rainfall and more droughts : productivity limited to 850 KG/ha/year

Adaptation of RAS with specific crops : sugar cane, leguminous, maize...during immature period

No banana.....

Global productivity is decreased due to poor prices (low incentive and no replanting, impact of leaf diseases, poor tapping and high frequency

In addition to low prices : CC constraints = a long range rubber killing process

Potential impact of climatic change

Reduction of annual rainfall (should be above 1200 mm/year)

Different rainfall patterns : more rains in the morning....

More dry months (2 to 4 months in South Sumatra,

More typhoons and wind

→ More potential competition between rubber and associated crops

→ Potential impact on rubber growth ?

→ Need for a balance between crops to limit competition

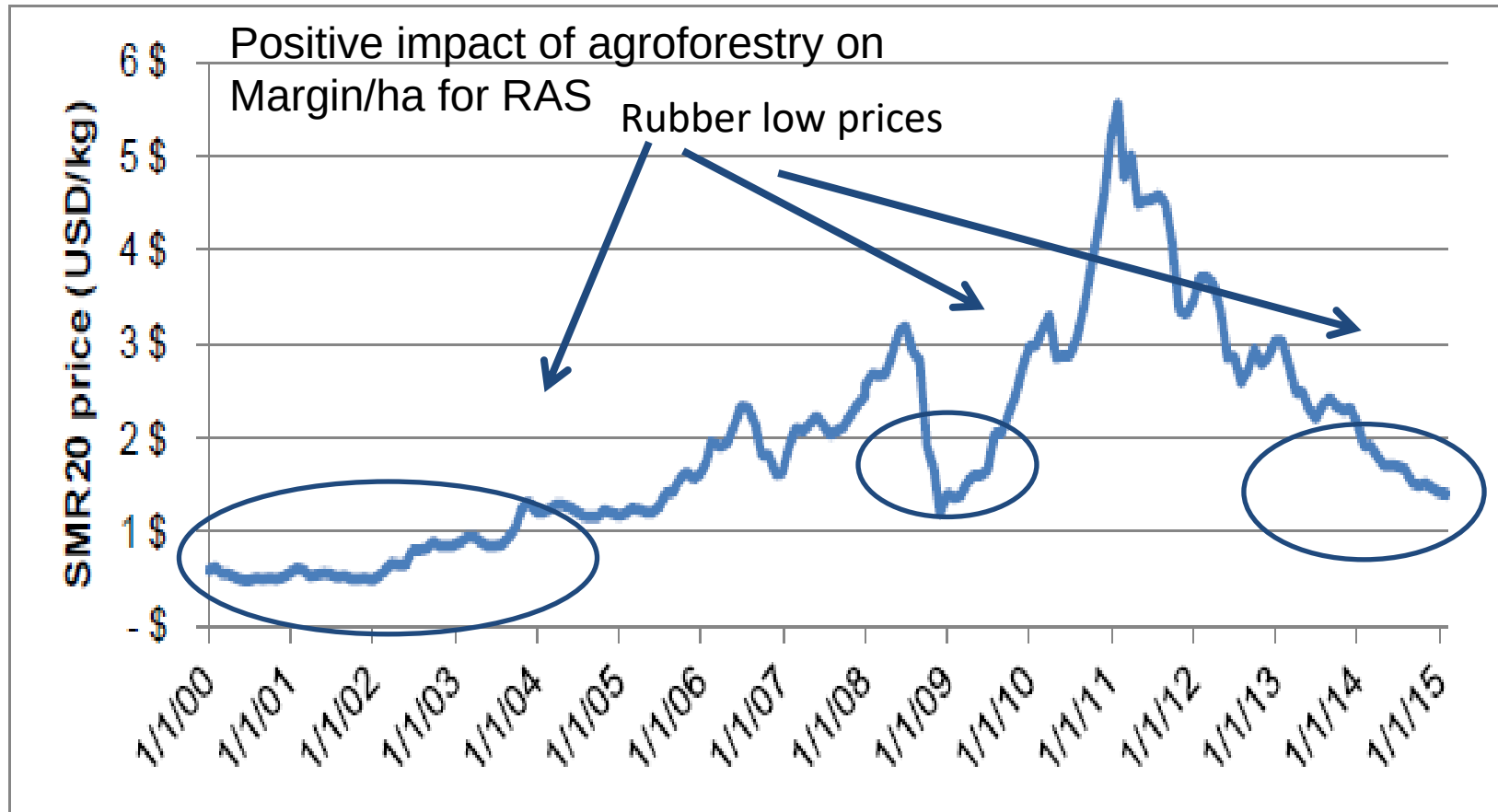
→ impact of CC on leaf disease and global rubber productivity ?

→ What is the best design if competition occurs

→ Single spacing or double spacing

IMPACT OF RUBBER PRICES

Thailand and Indonesia : farmers suffers from rubber prices volatility → high interest to income diversification and agroforestry practices = long term economic sustainability



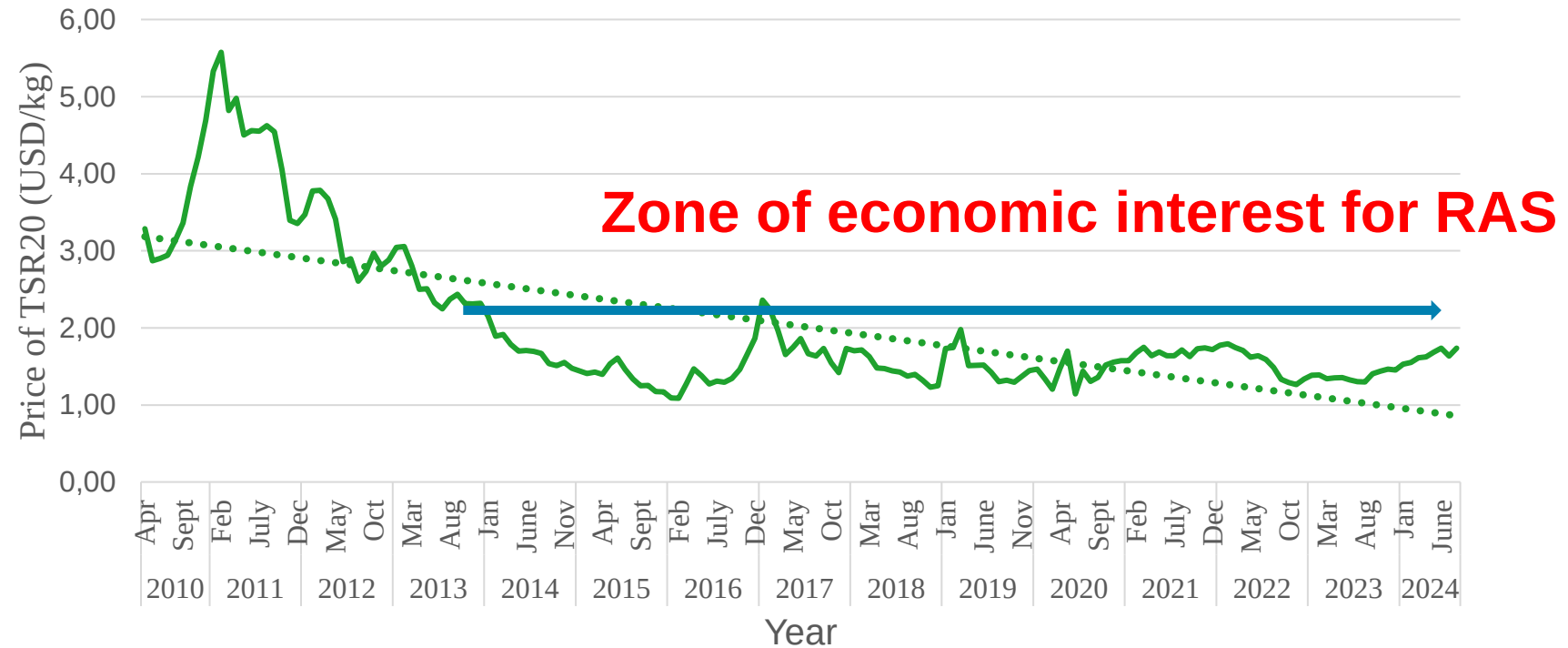
Rubber low prices
Rubber low prices

**Any link with
climatic
changes ???**

**No so far
But a real constraint
For further
Developemnt**

When is RAS economically interesting ?

Fluctuation of natural rubber price, 2010 - 2024



Source : SICOM, 2010 - 2023

Agroforestry patterns

Normal spacing (6 x 3 m or 7 x 3 m)

The « traditionnal way » for AF systems over the world (Indonesia, Thailand, Sri Lanka, India....)

Efficient with fruit trees and timber, plants for local markets (gnetum and fruits in Thailand) or export (cardamom in India).

- Lower productivity of associated trees but global gross margin/ha significantly improved.
- Interesting in situation with land scarcity, to reduce high dependance on rubber income and diversify agricultural income
- OK if the clone is adapted (RRIM 600 in Thailand) or if there is leaf disease (Indonesia, Sri Lanka)

Large/double spacing : from 8 to 25 m as inter-rows 1

1, 2 or even 3 lines of rubber at 2,2x 2,2 x2,5 and large inter=rows = double spacing

Systems developed in Sri Lanka and China with tea (Estates)

Trials in Gabon, Indonesia , Sri lanka and China (coffee, cocoa, tea, rice, banana...) as soon as the 1990's
Very few current development with smallholders

→ A real change in practices for farmers

Double spacing 2

→ Interesting if market is present and prices high to guaranty a profitable margin/ha.

Examples: coffee, cocoa, Durian, pineapple, banana....

→ Profit from associated trees should compensate significantly the expected lower production of rubber if tree planting density decrease until 400 trees/ha (minus 10 to 15 % of rubber production : an economic assessment is necessary

→ We need to identify which system is the best adapted locally

→ and if farmers are interested.....

Change of paradigm for farmers

Some examples in the world

systems with normal spacing : 3m x 6 m



CINNAMON



LEGUMES

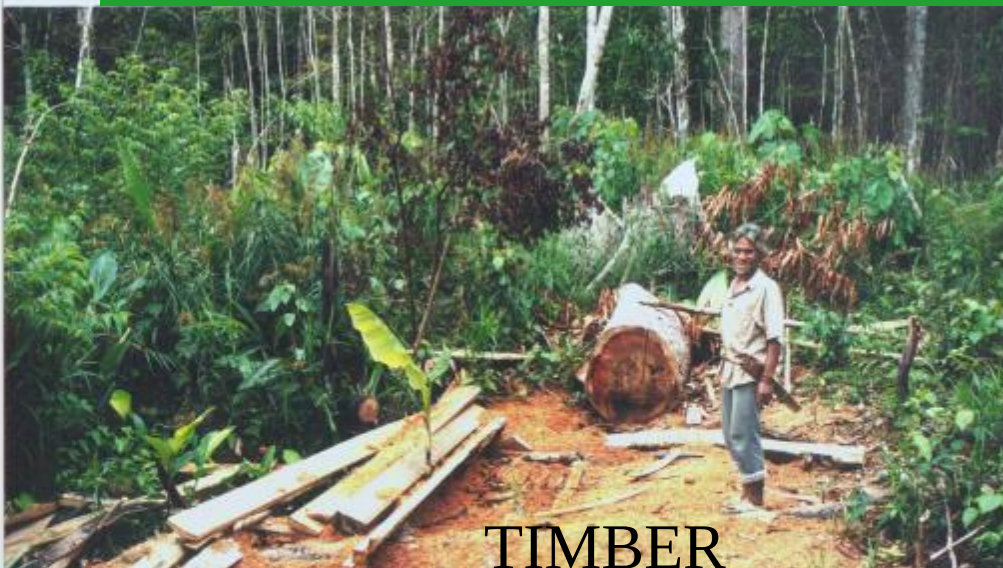


DURIANS



JENGKOL

A VERY HIGH VARIETY OF PRODUCTS



TIMBER



FERNS



MEDICINAL
PLANTS



Mushrooms



Damar resin



vegetables



**Rattan for
handicraft**



Fruits

Kalimantan /Indonesia : Rubber + Illipe nut (to produce an equivalent of cocoa butter) + **upland rice**



Sumatra/Indonesia

: anti erosion protection with flemingia congesta +
rubber + arec nut tree (bétel) + **upland rice**



West Sumatra/Pasaman/Indonesia: upland rice year 2





Photos on the Spots: RAS at both Station Research & Farmer' plot Trials: Annual Cash Crops

Station Research



Upland rice in betweenrubber



Chili in betweenrubber



Banana+Pineapple in betweenrubber



Maize in betweenrubber

Farmer' Plots



Groundnut+upland Rice in betweenrubber



Chili in betweenrubber



Pineapple betweenrubber



Watermelon betweenrubber

Some AF systems with intercrops in Thailand

Highly diversified
SAF systems
Linked to
a local context



Rubber-Pineapple



Rubber-Chili



Rubber-Corn



Rubber-Rice



Rubber-Papaya



Rubber-Banana

Marginal other systems in Thailand :

- Vegetables and /or tubers , pumpkin, cucumber, zucchini, melon, long beans, sweet potatoes
- marigold flower with irrigation; a market niche



Rubber + pakliang Thailand



Vietnam : intensification on limited land (low availability)

Rubber and groundnut



Rubber and rice



Rubber + Pineapple + fruits



Vietnam, 2000

Rubber + chine medicinal plants China



Recent association in North Sumatra in 2023 rubber with poor canopy due to leaf disease

- Rubber and coffe

Rubber and nutmeg





**Rubber
and sugar palm**



**Rubber
and
jemang /rattan**

**Rubber
and cacao**



- Rubber and Durian

Rubber and banana and Elephant grass with irrigation for export.... Vietnam 2024



Banana with irrigation



Elephant grass with irrigation

RAS in Thailand mature period

- RAS type developed by farmers themselves
- TIMBER TREES : neem tree or "thiem" (*Azadirachta excelsa*), "Thang" (*Litsea grandis*), a timber tree that grows naturally from natural regeneration in rubber fields, teak (*Tectonia grandis*), mahogany (*Switenia macrophylla*), "phayom": or white meranti (*Shorea talura*), "tumsao" (*Fragacs fragans*), *Acacia mangium*, rattan (*Calamus caesius* seems to be the most promising),
-
- FRUIT TREES : coffee (Robusta c), "Salak" (*Sallaca spp*), durian (*Durio zibethinus*), "longkong" (*Lansium domesticum*), "petai" (*Parkia speciosa* or Nita tree), "jack fruit" (*Artocarpus heterophyllus*), "cempedak" (*Artocarpus Integer*), "mangoustan" (*Garcinia dulcis*), and banana.
- + Bamboo, rattan, gnetum (legume leaf)

Three main types of permanent rubber AFS in Phatthalung province (Stroesser, 2015)



Type **Vegetable**
Mature rubber
associated with
vegetable species
(mainly Pakliang)



Type **Fruit**
Mature rubber
associated with fruit
and sometimes
vegetable species



Type **Timber**
Mature rubber
associated with
timber species

Rubber SAF in Thailand



Rubber +
Mangoustan +
Salak



Rubber + Tiam
+ Pakliang



Rubber +
Longkong



Rubber +
Mangoustan +
tiber trees



Jungle
rubber



Rubber + timber tree



Rubber + Salak

Rubber banana + coconut Thailand



Rubber + salak Thailand



Rubber + fruit trees Thailand



Durian

Various fruit trees

Net margin /ha for different agroforestry systems in southern Thailand in 2015

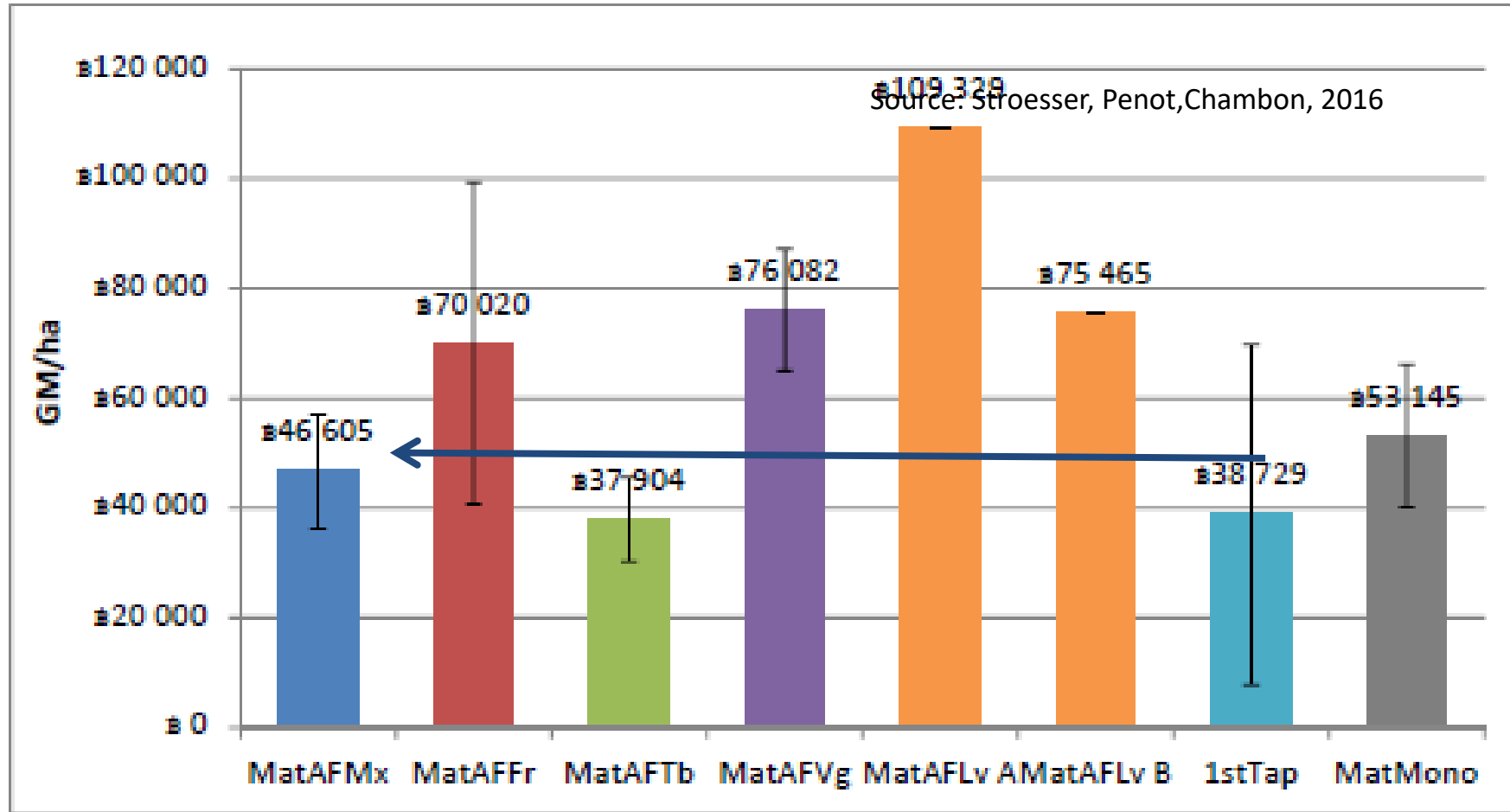
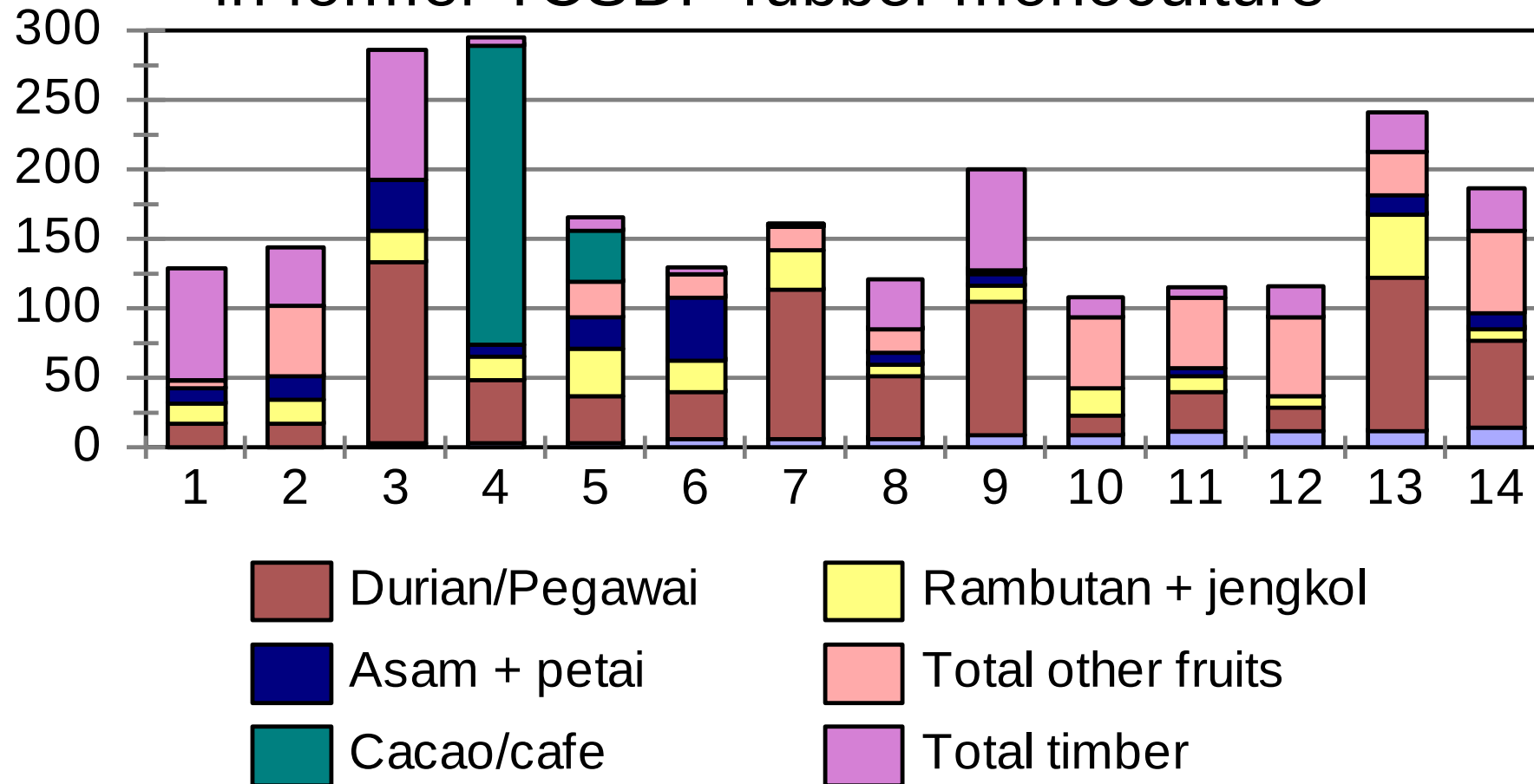


Figure 3 – Gross margin /ha for the AFS types and for a rubber monoculture

The situation in SRDP plots in 1997 with 110 to 300 associated trees /ha
In Indonesia

Associated tree distribution /ha /plot in former TCSDP rubber monoculture



Associated trees in RAS in Indonesia

Gaharu



**Timber
trees**

Pekawai



Rambutan



Acacia mangium



Oil palm

RAS smallholder's in Muaro Bungo District, Jambi (Rubis project surveys 2021/2022) With timber species



Rubber Agroforestry system with Meranti in Sepunggur Village, Muara Bungo (RAS project in 1996)



West Kalimantan
RAS 2 Year 11 (2007)
Fruit trees
local market for
jengkol and petai :



Sri Lanka : rubber + timber (Semesu) + cinnamon



Sri Lanka: rubber + pepper



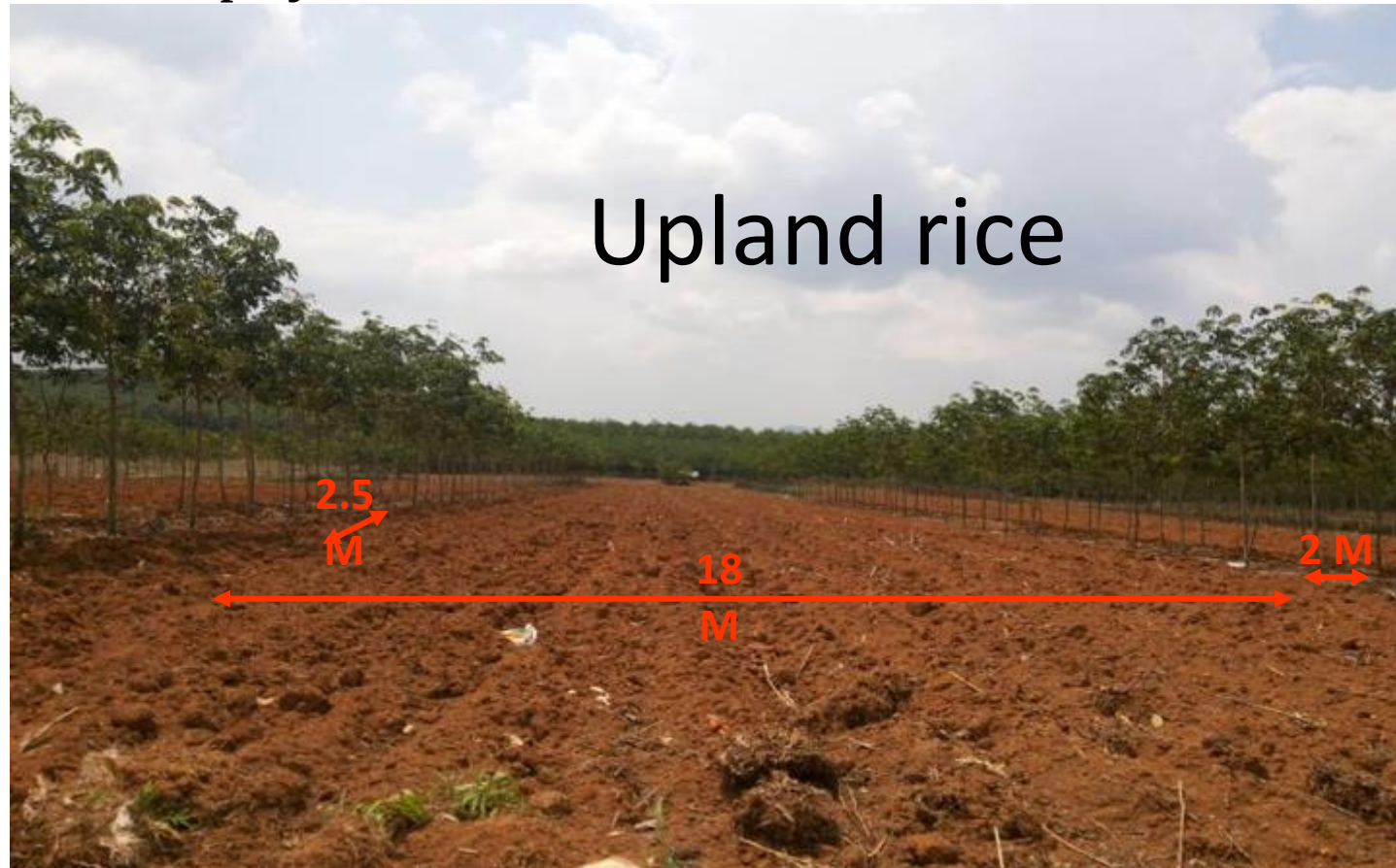
2 with double spacing :
3x2 X 8/12/15/18/25 m

IRRI rubber wide-row intercropping systems with banana and sugarcane at IRRI Sembawa immature period



- Productivity per hectare: benefit / cost ratio of 2.1
- Density of 400 trees/ha compared to 550 in standard plantation.

**Double row rubber intercropping system
separated by wide row spacing (18m x 2m) x
2.5m (Mr Sahuri, Sembawa RC, IRRI)
Rubis project 2022**



Using this wide row spacing, smallholders could cultivate intercrops between rubbers such as cash crops or other perennial crops

Myanmar Trials during immature period



Myanmar



Myanmar



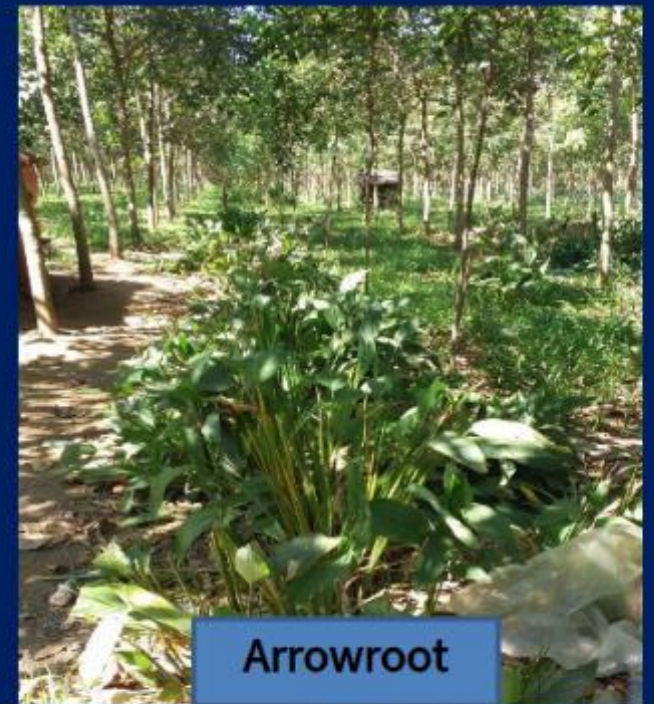
Myanmar



Myanmar



Sweet Potato



Arrowroot



Mangium



Betel Leaf

Rubber + sugarcane Myanmar and sri Lanka



**From year 3 to 6 with shading
which potential crops ?**

Rubber and lemongrass

Distillation unit for lemon
grass in intercropping



Other crop possibilities

- Elephant Foot Yam
- *Amorphophallus paeoniifolius*
- Taro (very rare)
- turmeric (*Curcuma longa* Linn.)
- cardamom



Or let weeds or covers on the soil (SAF
rubber/fruit in Thailand)

Impact on soils



Avoid noxious weeds :

Bracharia spp : severe impact on rubber growth (ex : Colombia)

Competition
for water



Indonesia : RAS in mature period with fruit and timber species



Arthocarpus



Teac, Double rows



Shorea, Double rows



Salacca



Fragrarea, Double rows



Timber



Durian



Pepper



Morinda, Double rows

Indonesia : source Gede Wibawa

Ivory Coast in 2010's: rubber and coffee

Spacing 18 m



Snoeck D., Lacote R., Keli Z.J., Doumbia A., Chapuset T., Jagoret P., Gohet E., 2013. Association of hevea with other tree crops can be more profitable than hevea monocrop during first 12 years. *Industrial Crops and Products*, 43 : 578-586.

Rubber + timber (teak and *Dalbergia* spp in Cambodia during mature period



Rubber + gaharu/Agar wood, Indonesia, 2022



Rubber and cinnamon in Sri Lanka



Rubber (400 trees/ha) and tea in Western Sri Lanka: No competition



Rubber and coffee Myanmar



Rubber and Durian Myanmar

2,5 x 4 x 2,5 (2 lines) x 25 m with pineapple



Rubber and banana and plantain (Africa) in 1992

Banana (Cote d'Ivoire)



Plantain (Gabon)



Rubber and tubers : what about cassava

- Cassava is not adapted as it is a plant from the same family of rubber and can favor white root disease if cropped more than 12 months
 - There is an obvious high competition for mineral nutrients between rubber and cassava.
 - Cassava can decrease rapidly soil fertility and cassava fertilization should be adapted.
- Very common in Northeast Thailand

**What is at stake in some countries with long dry season such as:
Myanmar, south Sumatra, Northeast Thailand...
and new areas with special climatic conditions or climatic change**

- Adaptation to specific climatic conditions : 3 to 5 months/year around 50 mm/month
- Sufficient rainfall in dry season ? -> competition ?
- Local or international market for associated product ? (spices...)
- Impact on rubber growth if any ?
- Impact of CC on leaf disease and rubber productivity

What do we expect from rubber agroforestry in double spacing ?

- Income diversification with specific crops not adapted to shading in normal spacing : specifically coffee, tea and cocoa and potentially annual crops (upland rice/indonesia)...others
- better yield under full sun : adapted for intensive systems (contracted agriculture ???)
- annual crops such as upland rice : good production (full sun + fertilizers) if rice price is sufficient
- necessity to make some prospective simulation select the right crops: rice , palawija, banana, pineapple, coffee, cocoa, other ????
- Economic simulation to compare double spacing AF systems and monoculture
- Survey farmers to see which systems are interesting for farmers: what are farmers priorities and strategies ?
- adapted to timber production for Estates: low maintenance in the double row with covercrops



Thanks for your attention